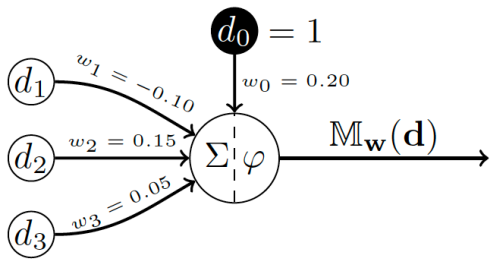
SYSC 5108 W26 Assignment 3

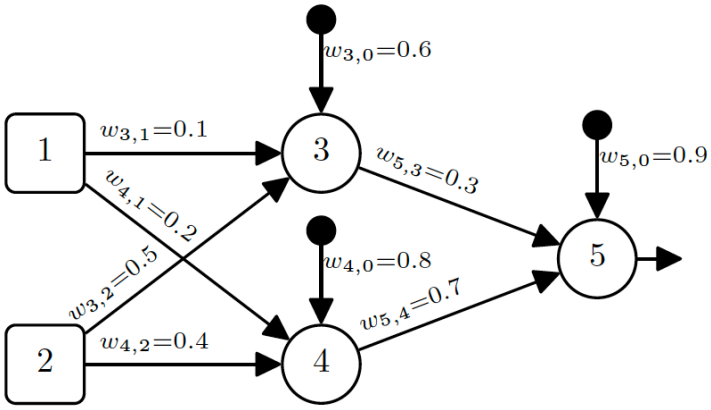
1. Maximum entropy is often used to select an a priori distribution when there is very little information about the RV. Consider the Bernoulli distribution $p(x) = \begin{cases} p, & x = 1 \\ 1-p, & x = 0 \\ 0, & \text{otherwise} \end{cases}$. Find its entropy $H(X)$ and prove analytically that $H(X)$ achieves its maximum when $p = \frac{1}{2}$.

2. The following image shows an artificial neuron that takes three inputs.

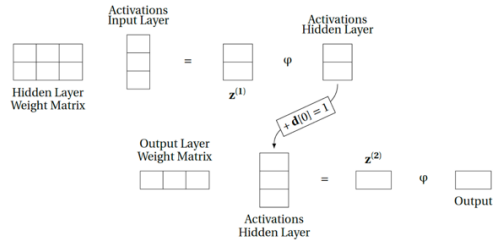


   (a) Calculate the weighted sum for this neuron for the input vector $\boldsymbol{d} = [0.2, 0.5, 0.7]$
   (b) What would be the output from this neuron if the activation function φ is a threshold activation with threshold $\theta = 1$?
   (c) What would be the output from this neuron if the activation function φ is the logistic function?
   (d) What would be the output from this neuron if the activation function φ is the rectified linear (ReLU) function?

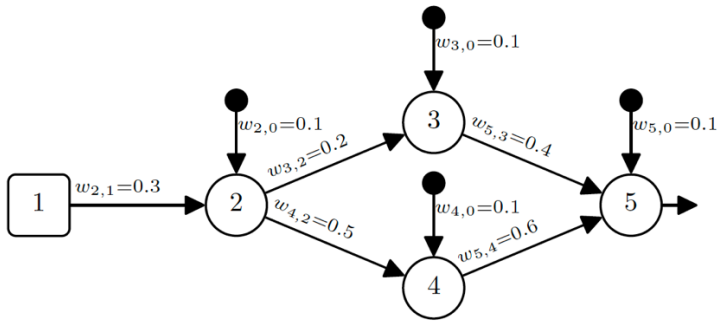
3. The following image shows an artificial neural network with two sensing (input) neurons (Neurons 1 and 2) and 3 processing neurons (Neurons 3, 4, and 5)



   (a) Assuming that the processing neurons in this network use a logistic activation function, what would be the output of Neuron 5 if the network received the input vector: Neuron 1 = 0.7 and Neuron 2 = 0.3?
   (b) Assuming that the processing neurons in this network use a ReLU activation function, what would be the output of Neuron 5 if the network received the input vector: Neuron 1 = 0.7 and Neuron 2 = 0.3?
   (c) The following image provides a template diagram for the sequence of matrix operations that our neural network would use to process the input vector Neuron 1 = 0.7 and Neuron 2 = 0.3. Assuming that the processing neurons in the network use a ReLU activation function, fill in the diagram with the appropriate weights, bias terms, weighted sum values, and activations.



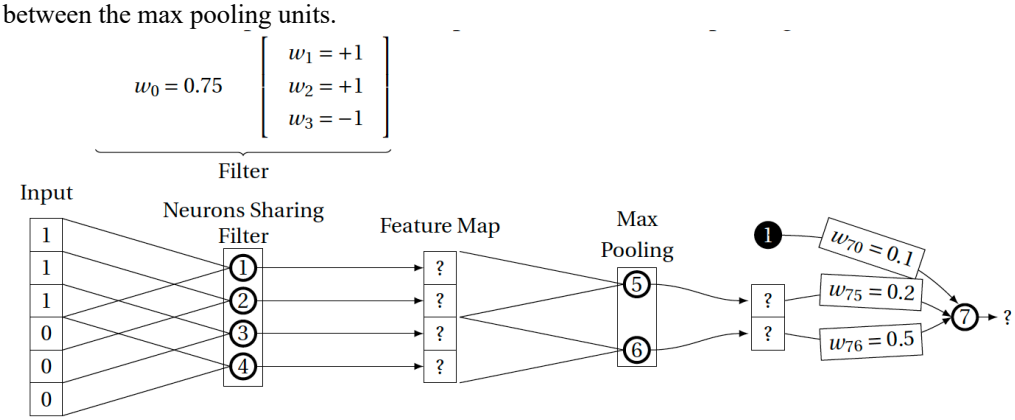
4. The following image illustrates the topology of a simple feedforward neural network that has a single sensing neuron (Neuron 1), three hidden processing neurons (Neurons 2, 3, and 4), and a single processing output neuron (Neuron 5).



Assuming that the processing neurons including output neuron are ReLUs, that the input to the network is Neuron 1 = 0.5 and that the desired output for this input is 0.9

(a) Calculate the output generated by the network in response to this input.
(b) Calculate the δ (error) values for each of the processing neurons in the network (i.e., δ_5, δ_4, δ_3, δ_2).
(c) Using the δ values you have calculated, calculate the gradient of the error of the network to changes in each of the weights of the network.
(d) Assuming a learning rate of $\alpha = 0.1$, calculate the updated values for each of the weights in the network. after the processing of this single training example.
(e) Calculate the reduction in the error of the network for this example using the new weights, compared with using the original weights.

5. The figure below illustrates a layer of a convolutional neural network that is processing a one-dimensional input. For ease of reference each of the neurons in the network has been labeled: 1; 2; 3; 4; 5; 6; 7. The architecture of the network consists of ReLUs that share a filter (Neurons 1; 2; 3; 4), followed by a sub-sampling layer containing two max pooling units (Neurons 5; 6), and then a fully connected layer containing a single ReLU (Neuron 7). The filter in the first layer has a window size 3, and a stride of 1. The max pooling units are 2to1, and there is no overlap between the max pooling units.



(a) What value will this network output?
(b) Assuming the target output for this input is 1, calculate the δ for each neuron in the network.
(c) Calculate the weight update for each weight in the filter: w_0;w_1;w_2;w_3.

6. Consider the example shown in Slide 28 of Chapter 4 with a feedforward network. Assume there is one hidden layer with 30 nodes and the output layer use multinoulli logistic regression. Find the number of learning parameters used. Next, consider Slide 20 of Chapter 5 with a CNN and multinoulli logistic output, calculate the number of learning parameters used.

7. In designing a convolutional layer for an image recognition data, what are the factors that affect the filter size, stride, number of channels?